

YOLO Detection TENSORRT (TRT) Model Inference Report

Backend Engine: TensorRT (TRT)

Phase 1: Configuration & System Metadata

Model Path	/workspace/Detection_Models/yolo26n.engine
Input Dimensions	640x640
Source Images Directory	/workspace/Images
Processed Images Output	/workspace/ENGINE_Output
OS	Linux 6.17.0-40-generic
Python Version	3.12.3
TensorRT Version	10.16.1.11
Inference Device	NVIDIA RTX A4000
CUDA Version	13.2
Model Architecture	YOLO (Detect)
Total Parameters	2400160
File Size	11.66 MB
Number of Classes	80
Precision Mode	FP32

Model Trained Class Names:

person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, traffic light, fire hydrant, stop sign, parking meter, bench, bird, cat, dog, horse, sheep, cow, elephant, bear, zebra, giraffe, backpack, umbrella, handbag, tie, suitcase, frisbee, skis, snowboard, sports ball, kite, baseball bat, baseball glove, skateboard, surfboard, tennis racket, bottle, wine glass, cup, fork, knife, spoon, bowl, banana, apple, sandwich, orange, broccoli, carrot, hot dog, pizza, donut, cake, chair, couch, potted plant, bed, dining table, toilet, tv, laptop, mouse, remote, keyboard, cell phone, microwave, oven, toaster, sink, refrigerator, book, clock, vase, scissors, teddy bear, hair drier, toothbrush

Phase 2: Inference Performance Metrics

Total Images Processed	5
Total Run Time	712.70 ms
Preprocess Latency	23.46 ms per image
Inference Latency	11.56 ms per image
Postprocess Latency	107.52 ms per image
Throughput	86.49 Images/sec
Peak Memory Usage	1496.62 MB

Phase 3: Detailed Predictions Summary

Image File	Detected Objects (Confidence)	Bounding Boxes (xyxy)
Cat_November_2010-1a.jpg.webp	cat (93.5%)	[208, 142, 892, 1219]
images.jpeg	dog (94.3%)	[87, 148, 422, 513]
images1.jpeg	zebra (92.6%)	[30, 173, 587, 416]
istockphoto-1189925691-612x612.jpg	bus (89.4%)	[69, 81, 583, 350]
tigermain.jpg	zebra (95.0%)	[668, 421, 1874, 1091]